Bolli Yaswanth Melki

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Professional Summary

Versatile engineer specializing in software engineering, with cross-domain expertise spanning full-stack web development, data science, and embedded systems. Smart India Hackathon competitor, Data Science Club core team member, and developer of diverse applications ranging from interactive websites to embedded systems.

Education

Marwadi University (NBA Accredited), Rajkot 2024 – 2028
B.Tech in Information and Communication Technology *CGPA: 6.78/10*

Work Experience

Airports Authority of India (AAI) – CNS Department, Rajahmundry, AP Jun 2026 – Jul 2026
CNS Intern

- Analyzed and monitored critical aviation communication and navigation infrastructure, gaining practical operational experience with VHF/HF transmitters, Doppler VHF Omni Range (DVOR), Distance Measuring Equipment (DME), and Instrument Landing Systems (ILS).Performed systematic hardware reliability testing, fault isolation, and root cause analysis on embedded communication devices to identify and resolve hardware-level failures.
- Collaborated with department mentors to analyze existing workflows and built a custom software solution designed to streamline their daily operations and improve efficiency.
- Evaluated power redundancy configurations, including Master-Slave and Load Sharing UPS setups, ensuring continuous and reliable power availability for mission-critical airport communication systems.Contributed to DRDO Technology Focus publication (Jan 2026) on health and habitability technologies for high-altitude defense environments.
- • Under the supervision of the CNS department supervisor, diagnosed and resolved a critical CMOS battery file load-up failure within the RCSI Rack by transferring site files via a Controller PC and rebooting the DVOR equipment to fully restore system functionality.

Projects

Project L – Freestanding x86 Operating System – GitHub 2026

• Built a custom freestanding 32-bit x86 operating system from scratch using C++, NASM Assembly, GCC, GNU Make, and QEMU, eliminating dependence on an existing operating-system kernel or distribution.

• Engineered the low-level boot and execution environment with direct hardware interaction, cross-compilation, linker configuration, and bare-metal system programming, establishing the foundation for kernel-level development.

LAD Engine – High-Throughput Local Analytics & Telemetry Engine – GitHub 2026

• Engineered a VS Code-native analytics engine capable of processing massive local datasets through chunked streaming and real-time telemetry, preventing client-side memory bottlenecks during heavy analytical workloads.

• Designed a Node.js–Python telemetry bridge with inter-process communication, memory-safe batch processing, and a custom Webpack build pipeline, packaging the system as a local .vsix extension for VS Code deployment.

Flow – GitHub Repository Intelligence & Portfolio Generator – GitHub 2026

• Built an AI-powered developer intelligence platform that analyzes GitHub repositories, commit history, technologies, and development patterns to reconstruct the technical story behind a project.

• Designed the workflow to transform raw repository activity into structured portfolio, resume, and project descriptions, reducing the manual effort required to document software engineering work.

Multi-Modal AI Damage Claim Agent – GitHub 2025

• Developed a multi-modal insurance claim processing agent combining image understanding, LLM reasoning, structured validation, and asynchronous processing to analyze damage evidence and generate claim intelligence.

• Integrated Gemini 1.5 Pro, DeepSeek, Pydantic, Pillow, Asyncio, and Pandas into an automated pipeline capable of processing visual and structured claim data through coordinated AI workflows.

Eqion – Engineering Formula Intelligence Workspace – GitHub 2025

• Built a web-based engineering formula workspace designed to provide students with a searchable, structured library of engineering equations and mathematical representations.

• Implemented responsive frontend workflows with Vanilla JavaScript and KaTeX, integrating Firebase Authentication and Firestore for user identity, persistent data, and cloud-backed application functionality.

Certifications

- **Web development: Zero to Hero by creating apps -** Udemy
- **Linux Fundamentals -** Intellipaat
- **Internship Certification -** AAI

Technical Skills

Languages & Scripting: Python, C++, Embedded C, JavaScript, OOP, Data Structures & Algorithms **AI/ML Frameworks:** PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Cuda, LangGraph, Model Optimization
Data & Signal Processing: DSP, DFT/FFT, Sampling, Convolution, Digital Filtering, Frequency-Domain Analysis, Audio Processing, Spectral Features, Feature Engineering
Embedded & IoT: ESP32, NodeMCU, Arduino, ARM Cortex, Firmware Development, UART, SPI, I2C
VLSI & Hardware: Hardware Trojan Analysis, Timing Analysis, KiCad, Proteus, Keil ,PCB Design
Software & Backend: React, Node.js, FastAPI, REST APIs, Firebase, Firestore, MongoDB, DBMS, SQL

Cloud & Deployment: Docker , Kubernetes – Intermediate

Tools: Git, GitHub, Linux, VS Code, Jupyter Notebook, Postman, Vite, NPM, Technical Documentation

Achievements & Leadership

- **ChatGenAI – College Fest:** Designed 30 high-end LLM-based problems and evaluated 50 participants based on their AI workflows, reasoning, and problem-solving approaches.
- **HackerRank Orchestrate Hackathon – Campus Crew:** Selected as a HackerRank Campus Crew member, contributing to the student developer community and hackathon ecosystem.
- **SIH 2025:** Participated in Smart India Hackathon 2025, developing an AI-driven solution under a national-level innovation challenge.
- **AMD Slingshot Ideathon:** Participated in the AMD Slingshot Ideathon, developing and deploying a functional web application under time constraints.
- **Data Science Club – Marwadi University:** Served as a Pseudo Committee Member, supporting the core team in organizing technical activities, student engagement, and data science initiatives.